CoreHive - Presentation Slides and Scripts

Slide 1: Current Situation in HR Management

Presenter: Imasha
Visuals to Include: Images showing cluttered paper files, Excel sheets, and a stressed HR worker.

Slide Content:

- Traditional HR processes rely heavily on manual paperwork.
- Data is scattered across fragmented tools (e.g., Excel for payroll, punch cards for time).
- High setup costs for traditional "On-Premise" Enterprise HR software.
- SMEs (Small and Medium Enterprises) cannot afford maintaining dedicated IT infrastructure.

English Script:
"Good morning, I'm Imasha. Let's look at the current HR situation. Today, many Small and Medium Enterprises use slow, paper-based methods and Excel for HR tasks. Existing 'On-Premise' Enterprise software is too expensive and requires dedicated IT servers, making it impossible for SMEs to afford proper HR systems."

Sinhala Script:
"සුභ උදෑසනක්. මම ඉමාශා. අදටත් ලංකාවේ ගොඩක් SMEs HR වැඩ කරන්නේ කඩදාසි සහ Excel මගින්. දැනට තියෙන Enterprise HR systems ගොඩක් ගණන් වගේම, ඒවා දුවන්න වෙනම servers සහ IT කට්ටිය ඕනේ. ඒ නිසා මේ පොඩි ආයතන වලට හොඳ HR system එකක් පාවිච්චි කරන්න අමාරු වෙලා තියෙනවා."

Slide 2: People Involved & Their Struggles

Presenter: Imasha
Visuals to Include: Icons of different roles (HR Manager, Employee, CEO) with a brief phrase under each.

Slide Content:

- **HR Managers:** Spend days calculating payroll, deductions, and tracking physical leave forms.
- **Employees:** Lack of self-service. Cannot easily check remaining leaves, payslips, or attendance records.
- **Business Owners:** Face high administrative costs and struggle to make data-driven decisions.

English Script:
"Three main groups face daily struggles here. HR Managers waste days manually calculating payroll and leave deductions, causing calculation errors. Employees lack a self-service portal to check leaves or past payslips, forcing them to visit HR personally. Finally, Business Owners face high administrative costs and lack clear HR analytics."

Sinhala Script:
"මෙතනදී ප්‍රධාන පාශර්ව තුනක් ප්‍රශ්න වලට මුහුණ දෙනවා. HR Managers ලට දවස් ගාණක් යනවා මැනුවලි පඩි සහ නිවාඩු හදන්න. Employees ලට තමන්ගෙ නිවාඩු හෝ පඩිපතක් බලන්න ක්‍රමයක් නෑ, HR එකටම යන්න ඕනේ. අන්තිමට Owners ලට මේකට ලොකු වියදමක් යනවා වගේම, හරියට data analyze කරලා තීරණ ගන්නත් අමාරුයි."

Slide 3: Core Identified Problems

Presenter: Imasha
Visuals to Include: A bulleted list with warning/alert icons emphasizing the pain points.

Slide Content:

- **Data Redundancy & Inconsistency:** Using multiple incompatible tools causes data mismatches.
- **Attendance Fraud:** Buddy-punching and inaccurate physical time records.
- **Scalability Blockers:** Manual systems cannot grow smoothly as the employee count increases.
- **Lack of Isolation:** Existing multi-company solutions don't ensure strong data privacy.

English Script:
"We identified four core problems: Data Redundancy from using separate tools like Excel and punch cards; Attendance Fraud like 'buddy-punching'; Scalability issues when manual systems fail to support growing staff; and severe Privacy Risks when using poorly isolated software for multiple companies."

Sinhala Script:
"අපි ප්‍රධාන ප්‍රශ්න 4ක් අඳුරගත්තා. වෙනස් tools පාවිච්චි කරද්දි data හරියට ගැලපෙන්නේ නැති එක. අනුන් වෙනුවට කාඩ් පන්ච් කරන Attendance Fraud එක. සේවකයෝ වැඩි වෙද්දි manual ක්‍රමය කරන්න බැරි වෙන Scalability ප්‍රශ්නය. අන්තිමටම, එකම system එකක් කොම්පැණි කිහිපයකට දෙද්දි data mix වෙන privacy ප්‍රශ්නය."

Slide 4: Requirements Collection Methods

Presenter: Savinda
Visuals to Include: Flowchart showing "Interviews -> Benchmarking (BambooHR/BetterHR) -> Survey -> Shortlisted Requirements"

Slide Content:

- **Stakeholder Interviews:** Interviewed SME owners to understand core HR struggles.
- **System Benchmarking:** Analyzed existing solutions like BambooHR and BetterHR, yielding a wide range of requirements.
- **Requirement Shortlisting:** Conducted a survey to prioritize features specifically needed within the Sri Lankan SME context.

English Script:
"Hello, I'm Savinda. For requirement gathering, we first interviewed SME owners. Next, we analyzed existing solutions like BambooHR and BetterHR, which gave us a wide area of potential requirements. Finally, to find out exactly what is genuinely needed within the Sri Lankan SME context, we conducted a survey to clearly shortlist and prioritize the most essential features."

Sinhala Script:
"මම සවින්ද. අපි මුලින්ම SME owners ලාව ඉන්ටවිව් කළා. ඊටපස්සේ BambooHR, BetterHR වගේ තියෙන systems analyze කරලා අපිට ඕනේ features ගොඩක් හොයාගත්තා. අන්තිමට, හරියටම ලංකාවේ කොම්පැණි වලට ගැලපෙන අත්‍යවශ්‍යම දේවල් මොනවද කියලා තෝරගන්න අපි survey එකක් කරලා requirements ටික shortlist කරගත්තා."

Slide 5: Identified System Functionalities & Roles

Presenter: Savinda
Visuals to Include: A User Personal hierarchy diagram: Super Admin -> Org Admin -> HR Staff -> Employee.

Slide Content:

- **System Admin (CoreHive Owners):** Approves new organization registrations, manages global platform statistics.
- **Organization Admin (SME Owners):** Buys subscriptions, manages billing, dynamic module activation.

- **HR Staff:** Manages departments, processes payroll, handles leave approvals.
- **Employees:** Personal dashboard access, dynamic QR generation, attendance tracking.

English Script:
"We defined four user roles for the system. The 'System Admin' manages new company registrations and global stats. The 'Organization Admin' acts as the SME owner, managing subscriptions and modules. 'HR Staff' handles daily tasks like payroll and leave approvals. Finally, the 'Employee' gets a self-service portal to generate attendance QRs and download payslips."

Sinhala Script:
"අපි ප්‍රධාන User Roles 4ක් හැදුවා. System Admin අපේ platform එකේ අයිතිකරුවෝ විදිහට කොම්පැණි approve කරනවා. Organization Admin subscription එක අරන් module තෝරනවා. HR Staff පඩි හදනවා සහ නිවාඩු අනුමත කරනවා. අන්තිමට Employee, එයාලගෙම dashboard එකට ලොග් වෙලා QR attendance මාක් කරන්න, නිවාඩු දාන්න සහ පඩි කොලේ ගන්න පුළුවන්."

Slide 6: Key Features (Based on Requirements)

Presenter: Savinda
Visuals to Include: Icons representing Cloud SaaS, QR code, Payhere, and Modules blocks.

Slide Content:

- **Multi-Tenant SaaS Model:** Pay-as-you-go subscription replacing heavy upfront software buying.
- **Secure QR Attendance:** Time-bound, dynamic QR codes preventing unauthorized check-ins.
- **Automated Payroll Engine:** Calculates exact formulas (Basic + Allowances - Leaves - Late Fines).
- **Dynamic Modularity:** Organizations can toggle extra features (e.g., Hiring Management) on and off.

English Script:
"Our key features address the identified problems directly. The Multi-Tenant SaaS Model offers a cheap pay-as-you-go subscription. Secure QR Attendance uses time-bound QR codes to stop buddy-punching. Our Automated Payroll Engine instantly processes late fines and leaves. Lastly, Dynamic Modularity allows companies to cut costs by turning off unused features."

Sinhala Script:
"අපේ ප්‍රධාන features 4ක් තියෙනවා. ලොකු වියදමක් නැති මාසිකව ගෙවන SaaS model එක. හොරට attendance දාන්න බැරි තත්පර ගානකින් expire වෙන Secure QR attendance එක. පරක්කු වුන වෙලාවල්, නිවාඩු ඔක්කොම

එකතු කරලා ඉබේම පඩි හදන automated payroll එක. අන්තිම එක තමයි තමන්ට අවශ්‍ය නැති features off කරලා සල්ලි ඉතුරු කරගන්න පුළුවන් Modularity පහසුකම."

Slide 7: High-Level System Architecture

Presenter: Shehan
Visuals to Include: A high-level Architecture Diagram linking Frontend (React) -> Backend API (Spring Boot) -> DB (MySQL) + Azure Blob.

Slide Content:

- **Architecture Pattern:** N-Tier, Multi-Tenant SaaS Application.
- **Frontend:** React.js, Vite, Tailwind CSS, Material UI (Decoupled Client).
- **Backend:** Spring Boot 3, Java 21, Spring Security (Stateless REST API).
- **Infrastructure:** MySQL Relational Database, Azure Blob Storage (Profile & Documents).

English Script:
"Hi, I'm Shehan. CoreHive uses a decoupled N-Tier architecture. We built the Frontend with React, Vite, and Tailwind CSS. It communicates securely via stateless REST APIs to a Spring Boot 3 Backend using Java 21. For data, we use a single MySQL relational database, and Microsoft Azure Blob Storage for handling multimedia files."

Sinhala Script:
"මම ශෙහාන්. අපි system එක හැදුවේ N-Tier architecture එකක් යටතේ. Frontend එකට React, Vite සහ Tailwind CSS පාවිච්චි කළා. Backend එක හැදුවේ අලුත්ම Java 21 සහ Spring Boot 3 වලින්. Database එකට MySQL පාවිච්චි කරනවා වගේම, පින්තූර සහ ලියකියවිලි save කරන්නේ Microsoft Azure Blob Storage කියන cloud සේවාවේ."

Slide 8: Database Design & Tenant Isolation

Presenter: Shehan
Visuals to Include: A simplified ERD showing the `Organization` table mapping to `Employee`, `Attendance`, and `LeaveRequest` via `organization_uuid`.

Slide Content:

- **Single-Database, Multi-Tenant Design:** All client organizations share the same database structure.
- **Tenant Isolation:** Every major table includes an `organization_uuid` foreign key.
- **Security Filter:** Spring Data JPA queries strictly filter by `organization_uuid` derived from the user's JWT token.
- **Scalability:** Very easy to maintain and deploy compared to Database-per-Tenant models.

English Script:
"To achieve SaaS multi-tenancy affordably, we use a 'Single-Database, Shared Schema' design. To prevent data leaks between companies, we enforce strict Tenant Isolation. Every major table has an `organization_uuid` foreign key. The backend extracts this UUID dynamically from the user's JWT token and filters every SQL query, guaranteeing complete data privacy."

Sinhala Script:
"අපි පාවිච්චි කළේ 'Single-Database' කියන ක්‍රමය. ඒ කියන්නේ ඔක්කොමලාගේ data තියෙන්නේ එකම database එකක් ඇතුලේ. ඒක වෙන් කරන්න අපි හැම table එකකටම 'organization_uuid' කියලා foreign key එකක් දුන්නා. කෙනෙක් ලොග් වුනාම එයාගේ JWT token එකෙන් ඒ UUID එක අරන් query එකටම දාලා ෆිල්ටර් කරන නිසා එක කොම්පැණියක විස්තර තව කොම්පැණියකට කීයටවත් පේන්නේ නෑ."

Slide 9: Core System Integrations & Data Flow

Presenter: Shehan
Visuals to Include: Flowchart demonstrating: Subscribe (PayHere) -> Webhook -> Auto-enable Modules.

Slide Content:

- **Payment Flow Integration:** Integrated with PayHere. Successful webhooks instantly update the `Subscription` table and unlock specific routing paths.
- **Attendance Architecture:**
  i. Employee generates secured JWT-encoded QR string.
  ii. Scanner validates token expiry & Org UUID.
  iii. Backend calculates Late Minutes vs. Office Start Time.
- **PDF File Creation:** jsPDF used alongside Axios to safely fetch Azure-hosted images for ID card generation.

English Script:
"Here are some core data flows. When an organization pays via PayHere, a webhook triggers our

backend to auto-activate their paid modules. For Attendance, the frontend generates a time-limited encrypted QR code. When scanned, the backend strictly validates the time against office hours to calculate late minutes automatically."

Sinhala Script:
"ප්‍රධාන data flows ගැන කිව්වොත්, PayHere එකෙන් සල්ලි ගෙව්වම එන webhook එකෙන් system එක auto active කරලා modules unlock කරනවා. Attendance දාද්දි හැදෙන encrypt වුන QR එක ස්කෑන් කරාම, එයා පරක්කුද නැද්ද කියලා backend එකෙන්ම කිරලා බලලා තමයි database එකට යවන්නේ. ඒ වගේම PDF අයිඩින්ටි කාඩ් හදන්න අපි වෙනම logic එකක් හැදුවා."

Slide 10: Implementation of Key Modules

Presenter: Shehan
Visuals to Include: Briefly annotated screenshots of the Application (e.g., The Subscription card view, The Payroll Table).

Slide Content:

- **Subscription Management Level:** Scheduled tasks check next-billing dates before applying plan downgrades.
- **Leave Management Flow:** Status pattern (Pending -> Approved/Rejected). Checked against annual allowance tables.
- **Payroll Execution:** Runs a heavy transactional calculation mapping Attendance (Absent days) + Basic Salary -> Deductions.
- **Module Disabling Logic:** Specific routing and UI components hide seamlessly if a module boolean returns false.

English Script:
"Our module logic handles real-world complexity. The Subscription module uses Spring Scheduled Tasks to apply plan downgrades only at the next billing cycle. The Payroll engine runs a complex database transaction, actively mapping attendance records and unapproved leaves to deduct from the basic salary. In the frontend, dynamic UI routing seamlessly hides any inactive modules."

Sinhala Script:
"අපේ ප්‍රධාන modules ගැන බැලුවොත්, කෙනෙක් plan එක downgrade කළොත් Spring Scheduled Task එකක් හරහා ඒක ඊළඟ මාසේ බිල ගෙවන දවසට විතරක් apply වෙන්න හදලා තියෙනවා. Payroll module එක ඒ මාසේ ආපු දවස් සහ පරක්කු වීම් අරගෙන, දඩ මුදල් කපලා තමයි අතට එන ගාන හදන්නේ. ඒ වගේම සල්ලි ගෙවලා නැති features වලට යන්න බැරිවෙන්න React UI එකෙන්ම හංගනවා."

Slide 11: Testing and Evaluation Approach

Presenter: Charith
Visuals to Include: A matrix or pyramid showing Manual Testing, Postman API Testing, and JUnit Testing.

Slide Content:

- **Primary Goal:** Ensure multi-tenant logic is secure and core HR calculations are accurate.
- **System Testing:** End-to-end flows involving React Frontend + Spring Boot Backend interaction.
- **Manual Testing (UAT):** Checked UI rendering, validations, and user flow for each role.
- **Backend API Testing:** Evaluated JSON payloads, HTTP Status Codes, and JWT Authorization via Postman.

English Script:
"Hello, I'm Charith. To ensure absolute data security and accuracy, we tested CoreHive rigorously in three ways. First, System Testing to verify end-to-end integration between React and Spring Boot. Second, Manual User Acceptance Testing where we logged in as different roles to validate the intuitive UI flow. Finally, Postman API Testing to ensure correct JSON responses and proper unauthorized blocking."

Sinhala Script:
"මම චරිත්. අපි ප්‍රධාන ක්‍රම 3කට මේක ටෙස්ට් කළා. පළවෙනි එක System Testing. Frontend, Backend ඔක්කොම එකට දුවන්න දීලා සම්පූර්ණ ක්‍රියාවලිය වැඩද බැලුවා. දෙක, Manual Testing. අපිම එක එක roles වලින් ලොග් වෙලා, UI එකේ අවුල් තියෙනවද කියලා බැලුවා. තුන්වෙනි එක API Testing. Postman මෘදුකාංගය හරහා backend APIs වලට කතා කරලා security එක සහ වැරදි requests block කරනවද බැලුවා."

Slide 12: Unit Testing (Backend Logic Validation)

Presenter: Charith
Visuals to Include: A screenshot of a successful JUnit Test Run from the IDE (showing Green checkmarks for AuthServiceTest).

Slide Content:

- **Frameworks Used:** JUnit 5 and Mockito.
- **Isolated Testing:** Business logic tested without altering the real database state.

- **Key Scenarios Tested:** Login success/failure, Invalid Passwords, Fetching Active Plans.
- **Exception Handling:** Ensuring proper backend crashes and error messages when edge cases occur.

English Script:
"To guarantee our core backend logic never fails, we wrote Unit Tests using JUnit 5 and Mockito. By mocking our database responses, we isolated and tested Service classes like AuthService and OrganizationService. We successfully tested edge cases, like submitting invalid passwords, ensuring the server simply throws clean error messages instead of completely crashing."

Sinhala Script:
"System එකේ logic එක වරදින්නේ නෑ කියලා සහතික කරන්න අපි JUnit 5 සහ Mockito පාවිච්චි කළා. ඇත්ත database එකට කතා කරන්නේ නැතුව, Mockito හරහා බොරු data දීලා අපේ Service classes වෙනම ටෙස්ට් කළා. උදාහරණයක් විදිහට වැරදි password එකක් දුන්නොත් හරියටම error message එක එනවද, මුළු backend එකම crash වෙනවද කියලා අපි කෝඩ් ලියලම ටෙස්ට් කරලා තහවුරු කළා."

Slide 13: Evaluation Details against Requirements

Presenter: Charith
Visuals to Include: A short checklist graphic showing goals vs. achieved logic. (e.g., Role Access -> Passed, Calculation -> Passed).

Slide Content:

- **Role-Based Routing Verified:** Successfully blocked lower-tier users from Admin dashboards.
- **Tenant Data Security Verified:** Confirmed an organization cannot query another tenant's arrays.
- **Performance Checked:** JWT authentication execution generates negligible overhead.
- **Functional Accuracy:** Leave balance deduction and payroll sums manually calculated to cross-verify system output.

English Script:
"Our final evaluation confirmed all user requirements were met safely. Role-based routing flawlessly blocked employees from accessing HR dashboards. Most importantly, Tenant Data Security succeeded—injecting false company UUIDs into APIs was correctly rejected. Finally, we manually calculated sample salaries and leave balances using a physical calculator, and our automated Payroll JSON matching them perfectly."

Sinhala Script:
"අවසාන ඇගයීමේදී අපි බැලුවා අවශ්‍යතා ඔක්කොම ඉටු වෙලාද කියලා. Roles හරියටම වැඩ කරනවා, Employees ලාට අනිත් අයගේ විස්තර බලන්න බෑ. වැදගත්ම දේ වෙන කොම්පැණියක ID එකක් දීලා හොරෙන් තොරතුරු ගන්න හැදුවාම backend එක ඒක ප්‍රතික්ෂේප කරා. ඒ වගේම අපි අතින් කැල්කියුලේටරයක් අරන් පඩි හදලා, system එකෙන් ආපු අවසාන ගානත් එක්ක සසඳලා බලද්දි ඒ දෙකම හරියටම සමාන වුණා."

Slide 14: System Weaknesses & Limitations

Presenter: Charith
Visuals to Include: Alert or minor caution icon next to bullet points.

Slide Content:

- **Hardware/Network Dependency:** QR attendance requires strict internet connectivity at the office premise.
- **Lack of a Native Mobile App:** Current mobile experience is restricted to web browsers, limiting background GPS interaction.
- **Complex Customization Constraints:** Completely unique company-specific payroll formulas might require direct code intervention.

English Script:
"Evaluating our system's weaknesses, the biggest limit is strict network dependency. If office WiFi fails, QR check-ins are blocked. Additionally, we lack a Native Mobile App. While fully responsive on phone browsers, a native app would better utilize background GPS. Furthermore, highly customized company salary formulas cannot easily bypass our standardized payroll engine."

Sinhala Script:
"අපේ system එකෙත් දැනට දුවර්ලතා තියෙනවා. ලොකුම ප්‍රශ්නය මේක ඉන්ටනෙර්ට් මත යැපෙන එක. ඔෆිස් එකේ wifi නැති වුණොත් QR කෝඩ් එක ස්කෑන් කරලා attendance දාගන්න බෑ. අනිත් එක මේකට Native Mobile app එකක් නැති එක. ඇප් එකක් තිබ්බා නම් අපිට GPS location ලේසියෙන් ගන්න තිබුනා. වගේම, සමහර ආයතන වල තියෙන අමුතුම පඩි නීති මේ system එකේ දැනට හදාගන්න අමාරුයි."

Slide 15: Conclusion and Future Work

Presenter: Charith
Visuals to Include: Forward-looking icons (Rocket, Mobile App, AI Brain).

Slide Content:

- **Conclusion:** CoreHive successfully provides an affordable, secure, and automated HR SaaS platform solving physical data fragmentation for SMEs.
- **Future Work - Offline Support (PWA):** Addressing network issues via Progressive Web Apps caching.
- **Future Work - Native Application:** Flutter/React Native app enabling Geofenced GPS check-ins.
- **Future Work - AI Analytics:** Introducing Resume parsing and predictive employee turnover models.

English Script:
"In conclusion, CoreHive successfully standardizes SME HR operations affordably. To tackle our weaknesses, future work includes upgrading to a Progressive Web App to allow offline check-ins. Second, building a Flutter Mobile App to introduce GPS 'Geofenced' attendance. Finally, we plan to inject AI to automate resume parsing and provide predictive alerts for high-turnover employees. Thank you."

Sinhala Script:
"අවසාන වශයෙන්, CoreHive හරහා පොඩි ආයතන වලට කඩදාසි වලින් තොරව අඩුවියදම් HR SaaS platform එකක් අපි නිමර්ාණය කළා. අනාගතයේදී ඉන්ටනෙර්ට් ප්‍රශ්නෙට විසඳුමක් විදිහට මේක PWA එකක් විදිහට දියුණු කරලා offline වැඩ කරන්න හදනවා. ඒ වගේම Flutter වලින් Mobile app එකක් හදලා GPS මගින් ඔෆිස් එකේදී විතරක් attendance වැටෙන 'Geofencing' ගේන්න හිතන් ඉන්නවා. අන්තිමට AI හරහා CV කියවන එක වගේ දේවල් ඉස්සරහට එකතු කරනවා. ස්තූතියි!"